

The Importance of Medication Traceability in Hospital Logistics: Technology in the Service of Patient Safety

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Abstract. *This study presents the development of a medication traceability system for public and private hospitals, aimed at preventing losses and improving supply chain efficiency. The solution integrates hardware and software, using Arduino microcontrollers in C++ to read RFID tags linked to each medication. Data is sent to a web-based application that enables real-time tracking and logs each checkpoint. Simulated tests showed reduced losses, increased logistical transparency, and improved patient safety. The system reinforces the role of tracking technologies and automated control in strengthening information security and hospital resource management.*

1. Introduction

In recent decades, technological advances have triggered a profound transformation in the healthcare sector, impacting everything from clinical diagnosis to hospital management. The integration of technologies such as the Internet of Things (*IoT*), Artificial Intelligence (*AI*), and automation systems has enabled a significant improvement in the way healthcare services are delivered to the population. In the hospital context, this technological revolution not only optimizes internal processes but also enhances security, efficiency, and transparency in the management of essential resources, such as medications (Silva, Souza, & Andrade, 2021).

Artificial Intelligence, in particular, has played a strategic role in the analysis of hospital data, inventory tracking, and the detection of operational anomalies. Combined with technologies such as *RFID* (Radio-Frequency Identification), *AI* enables the implementation of intelligent systems for the traceability of pharmaceutical supplies, which not only control the flow of medicines but also assist in identifying losses, deviations, or thefts [Ferreira, Lima, & Barbosa, 2023; Almeida, Lopes, & Silva, 2024]. *RFID*, in turn, consists of tags and readers (*transceivers*), operating through radio-frequency waves to identify and track objects automatically and without direct physical contact [Want, 2006].

The issue of drug theft within Brazilian hospitals has generated significant financial and social impacts. Recent studies indicate an increase in records of unexplained losses of supplies, which directly compromises patient safety and the continuity of treatments [Costa et al., 2022]. Moreover, the lack of effective control over the medication life cycle hinders audits, impedes traceability in cases of adverse events, and opens loopholes for internal fraud [Gomes, Pereira, & Martins, 2023]. However, the adoption of *RFID*

technology also requires attention to information security concerns, as unprotected systems are vulnerable to threats such as eavesdropping, tag cloning, and denial-of-service attacks [Mubarak, 2011; Ouafi, 2012].

The literature indicates that attacks on *RFID* systems can be classified as active or passive. Active attacks involve data interception, manipulation, and device spoofing, whereas passive attacks include unauthorized eavesdropping and data capture [Mubarak, 2011]. Therefore, the implementation of measures such as encryption, mutual authentication, and secure key management is essential to ensure the integrity of transmitted data [Ouafi, 2012].

In this context, the development of integrated solutions combining low-cost hardware (such as Arduino), RFID sensors, and software systems with real-time monitoring capabilities emerges as a viable and scalable technological response to this challenge. The proposed approach demonstrates the efficiency of the equipment itself, showing that radio-frequency tracking can be applied even in smaller-scale infrastructures. The RFID antennas, configured for short-range operation, proved capable of accurately identifying and recording the movement of medications, maintaining data consistency and reinforcing transparency in hospital logistics operations.

This practical evidence confirms that the technology can be adapted to different hospital realities, ensuring traceability and security even in contexts with structural limitations. Furthermore, the integration between hardware and the web system guarantees real-time visibility of each stock movement, strengthening automated control, data integrity, and institutional trust in operational processes. Thus, the use of intelligent traceability solutions is consolidated as an essential tool to protect lives and ensure the ethical and efficient allocation of healthcare resources [Moura, Costa, & Martins, 2023].

2. Theoretical Framework

2.1. *RFID* and Its Applications in Healthcare

Radio-Frequency Identification (RFID) technology has emerged as an efficient solution for real-time tracking and identification of objects without the need for physical contact or a direct line of sight. Its application has expanded to several sectors, including healthcare, where it enables inventory control, equipment tracking, and, more recently, the management of medications within hospital units [Want, 2006; Silva, Souza, & Andrade, 2021].

In the present study, RFID was employed in conjunction with an experimentally built antenna, designed to operate on a reduced radio-frequency scale, demonstrating the feasibility of the technology in hospital contexts without the need for high-cost equipment. The combination of the handcrafted antenna, RFID reader, and Arduino microcontroller enabled communication between tags and the monitoring system, showing that it is possible to apply traceability accurately and securely even in simplified structures. This integration reinforces the potential of RFID as an accessible and adaptable tool for hospital environments [Ouafi, 2012; Almeida, Lopes, & Silva, 2024].

2.2. Medication Traceability in Hospitals

Strict control over the life cycle of medications within hospitals is an increasing concern, especially given the rise in cases of theft, loss, and diversion of high-value supplies. Stud-

ies indicate that failures in internal logistics processes have contributed to the increase in such occurrences, compromising not only the financial resources of institutions but also patient safety [Costa, Oliveira, & Santos, 2022].

The use of RFID technology with the antenna developed in this project demonstrated that it is possible to track medications at different points within the hospital, ensuring greater control and visibility over the flow of supplies. The integrated system—composed of a reader connected to the antenna, a communication backend, and a web-based frontend interface—allowed each registered tag to be linked to a specific medication, ensuring transparency of movements and reliability of displayed information. This application confirms that RFID can serve as the foundation for innovative and sustainable solutions in hospital management [Ferreira, Lima, & Barbosa, 2023].

2.3. Information Security in Hospital *RFID* Systems

Despite its operational benefits, the use of RFID in hospital environments also raises concerns regarding information security. Unprotected systems can be targeted by attacks such as eavesdropping, data interception, tag cloning, and denial-of-service attacks [Mubarak, 2011; Ouafi, 2012]. Such vulnerabilities pose significant risks, particularly when dealing with controlled or high-cost medications.

To mitigate these issues, best practices include implementing mutual authentication protocols, using encryption in communications between tags and readers, and adopting access control policies for database management [Almeida, Lopes, & Silva, 2024]. In the context of the developed system, these measures were considered from the early design phase, ensuring that communication between the antenna, reader, and server occurs securely. Thus, adopting these practices is essential to ensure the integrity, confidentiality, and availability of information generated by the hospital traceability process.

3. Methodology

3.1. Development Strategy

The development of the *Passou Aqui* system followed a hybrid approach, combining *Design Thinking* principles with *agile software development* practices. This allowed for a deeper understanding of hospital workflows and the delivery of rapid iterations focused on medication traceability. During the discovery phase, interviews were conducted with healthcare and IT professionals to map critical logistics points; in the **design** phase, the overall system architecture was defined, integrating *hardware* (*ESP32/Arduino with RFID module*), *backend* (*Django with PostgreSQL*), and *frontend* (*Next.js with React and Tailwind CSS*), along with prototypes of the web interfaces (see **Figure 1** and **Figure 2**).

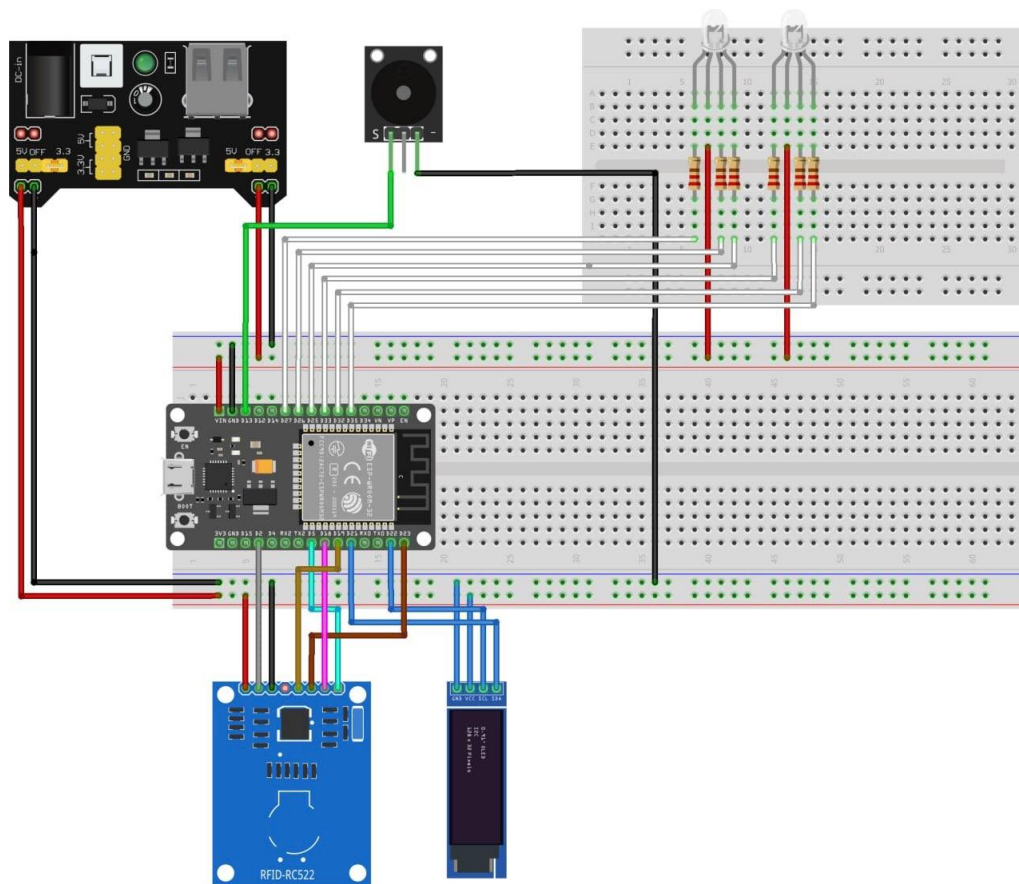


Figure 1. Architecture of the medication reader.

Source: The authors.

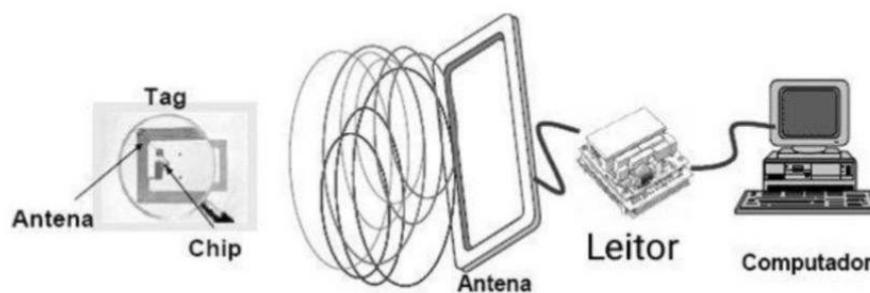


Figure 2. Illustration of a generic RFID system.

Source: PUHLMANN, 2015.

Finally, the development stage emphasized module integration and experimental hardware validation. In addition to assembling the MFRC522 reader connected to the microcontroller, a **handcrafted copper coil antenna** was built to operate on a reduced scale and test the practical feasibility of tracking in hospital environments. Tests were conducted in both virtual environments (*Tinkercad*) and physical ones (*ESP32 + MFRC522*),

with visual verification via *LEDs* and reading transmissions sent through *HTTP/JSON* to the backend. These tests confirmed the communication between tag, reader/antenna, and monitoring system—an essential condition for reliable traceability (see **Figure 3**).

The validation phase focused on assessing performance (latency, read consistency) and robustness. Tests were applied to investigate the influence of the **copper coil diameter** on the sensitivity and reading range of the *MFRC522*, adjusting the antenna geometry to improve resonance and signal efficiency in low-infrastructure scenarios. Simulated intermittent connectivity situations were also performed to validate batch transmission and temporary local storage.

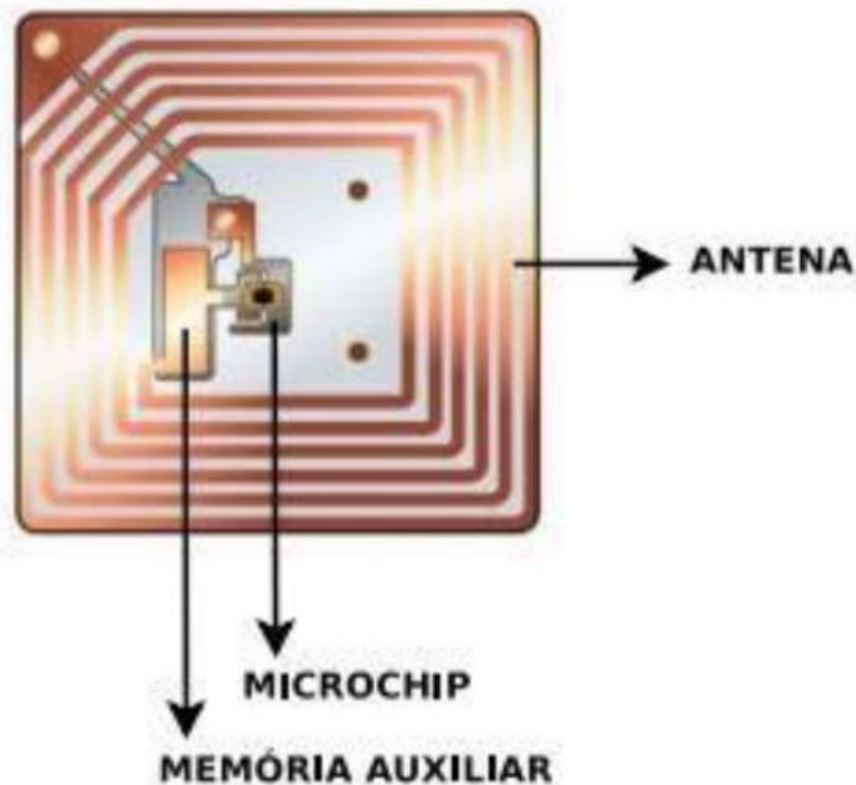


Figure 3. Structure of an RFID tag.

Source: MOTA, 2015.

3.2. System Architecture and Components

The system was designed in three layers: *hardware*, *backend*, and *frontend*. In the *hardware* layer, the *MFRC522* reader coupled to the microcontroller (ESP32/Arduino) interfaced with the *RFID tags*, while the **copper coil** functioned as an adjustable handcrafted antenna for field testing. This configuration allowed for the evaluation of RFID applicability using low-cost, small-scale components, demonstrating that a homemade antenna can meet traceability requirements when properly calibrated.

The *backend* was implemented using the **Django REST Framework** and **PostgreSQL**, responsible for validating *UIDs*, recording logs with timestamps, and exposing

APIs consumed by the *frontend*. The *frontend*, developed with *Next.js*, *React*, and styled with *Tailwind CSS* (with *Mantine UI* support during testing), displayed real-time readings, history, and filters by *tag*, *status*, and *user*. The modular architecture facilitates replicability and future replacement of components (e.g., commercial antennas) without substantial modification to the *backend* or *interfaces*.

During integration testing, antenna parameters (number of turns and coil diameter) were recorded and correlated with *MFRC522* reading metrics to identify configurations that maximized reliability without relying on professional-grade equipment.

3.3. Validation, Security, and Communication

Data validation and processing were designed with a focus on information security and integrity. The *backend* applies automated validations to ensure the authenticity of the tag's *UID*, correct reader registration, and timestamped logging. *Token-based authentication (JWT)* was implemented to ensure that only authorized devices can communicate with the system. In parallel, communication security measures were considered from the early stages of system design to preserve confidentiality and data integrity.

Regarding *hardware–software* communication, the *MFRC522* reader, coupled with the handcrafted antenna, captures the tag *UIDs* and transmits the data to the microcontroller, which packages the readings into *JSON* and sends them to the *backend* via *HTTP*. In scenarios with limited connectivity, the device stores records locally and performs batch transmission once the network becomes available, preventing data loss. The **master–slave** principle between reader and tag was maintained as the operational foundation (see **Figure 4**), complemented by *server-side* controls to prevent record duplication through *timestamps* and *UID* verification.

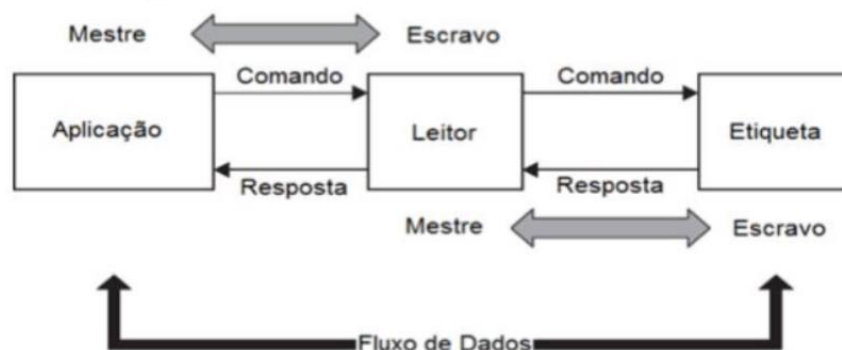


Figure 4. Master–slave principle used in RFID readers.

Source: MONTALVÃO, 2016.

In the experimental tests, common vulnerabilities in *RFID* systems (eavesdropping, cloning, interception) were considered, and mitigation practices were incorporated into the development workflow. These measures ensure that the solution—despite using low-cost antennas and readers—operates with adequate security levels and provides reliable data for hospital management.

4. Development and Implementation

The development of the *Passou Aqui* system originated from the need to improve the control and traceability of medications in hospitals. The absence of consistent records and automated verification mechanisms hindered auditing processes and the identification of losses. The proposed solution was structured in integrated modules, encompassing everything from the physical reading of the tag to the real-time visualization of information in a web environment, ensuring a continuous and reliable data flow.

During the construction phase, a modular architecture was defined, based on three layers: hardware, *backend*, and *frontend*. The hardware, consisting of an ESP32 microcontroller with an MFRC522 RFID reader, was responsible for identifying tags and sending information to the server. Experimentally, a handcrafted copper coil antenna was tested to measure the reading range and stability under different field conditions, demonstrating that the system can operate in low-infrastructure scenarios without compromising traceability.

The *backend*, developed with the *Django REST Framework* and a *PostgreSQL* database, implemented authentication and *UID* validation routines, duplicate record control, and timestamp logging. The *frontend*, built with *Next.js* and *React*, was responsible for dynamically and responsively displaying information, including filtering features, movement history, and medication status tracking. This integration resulted in an intuitive interface that expanded the operators' visibility over hospital supply movements.

To ensure legal compliance, the project followed the guidelines of the **General Data Protection Law (LGPD)**, since the processed information involves sensitive health-related data. Mechanisms such as token-based authentication (*JWT*), role-based access control, and encryption of data in transit were implemented, ensuring confidentiality and traceability of all operations within the system.

The application was tested in both virtual and physical environments, employing simulations in *Tinkercad* and the ESP32 + MFRC522 setup. These tests confirmed the stability of communication between the reader, antenna, and system, validating the performance and consistency of transmissions. The system demonstrated the capability to maintain control and traceability even during temporary connection loss, performing deferred data transmission from locally stored records.

Finally, the practical results evidenced the feasibility of using RFID technology as a support tool for hospital management. The *Passou Aqui* system proved effective in reducing errors, increasing transparency in medication flow, and strengthening information security. Its scalable architecture, compliant with LGPD principles, allows replication in different hospital contexts, consolidating RFID as an innovative and accessible solution for the healthcare sector.

5. Results

The application developed for the *Passou Aqui* system features an intuitive and functional interface designed for medication management in hospital environments. From the authentication screen, shown in Figure 5, professionals such as nurses and pharmacy

staff access the system through specific credentials. After logging in, the user is redirected to the *homepage*, shown in Figure 6, where real-time medication tracking can be initiated or the complete movement history can be consulted.

Medication registration takes place in the management section, illustrated in Figures 7 and 8, where it is possible to record information such as the drug name, RFID tag number, stripe type, and activity status. Once registered, medications are displayed in a simplified view for quick identification and can later be accessed in detail, as shown in Figure 9.

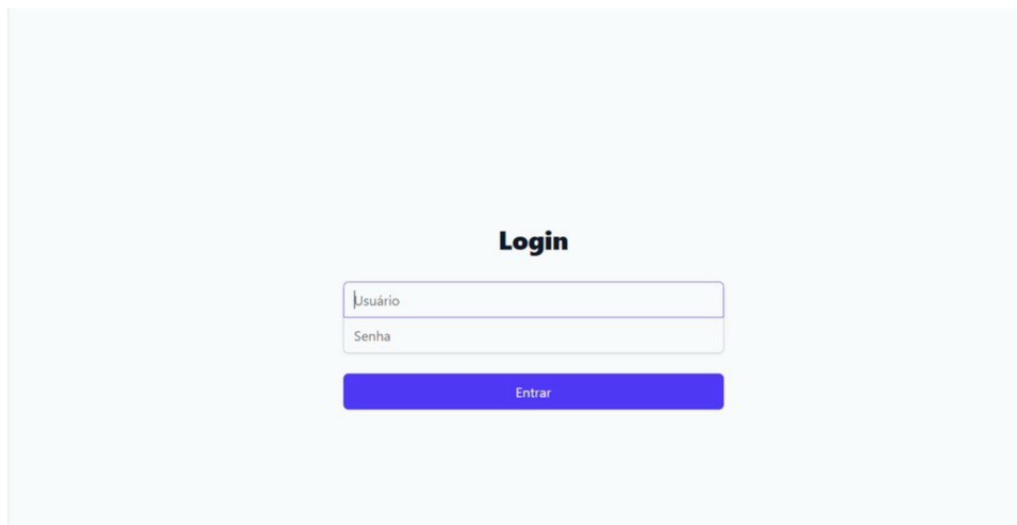


Figure 5. Website login page.

Source: The authors.

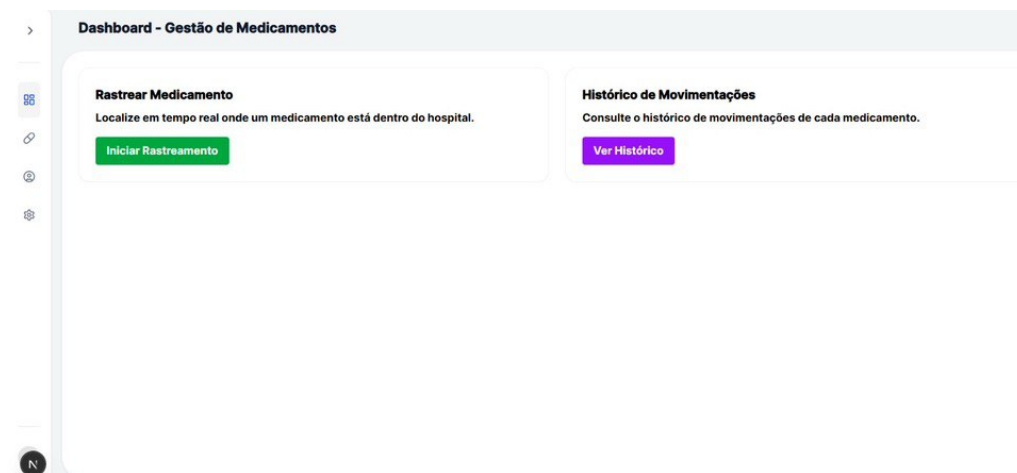


Figure 6. Website main page.

Source: The authors.

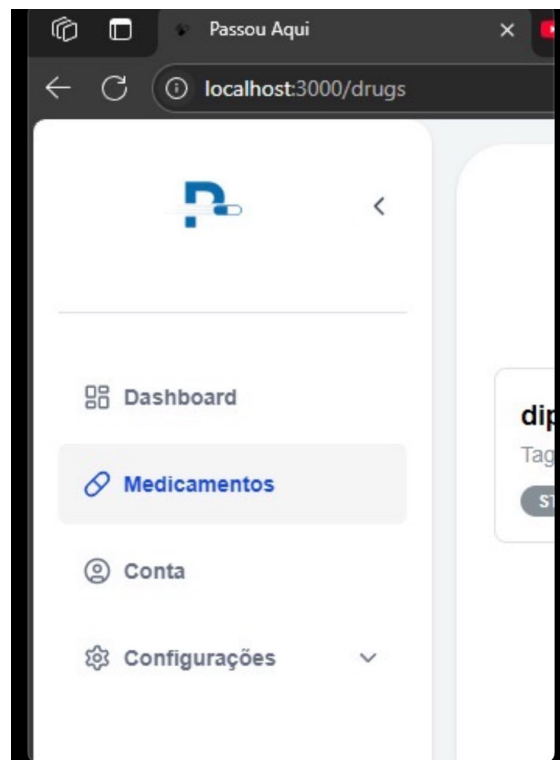


Figure 7. Medication catalog page.

Source: The authors.

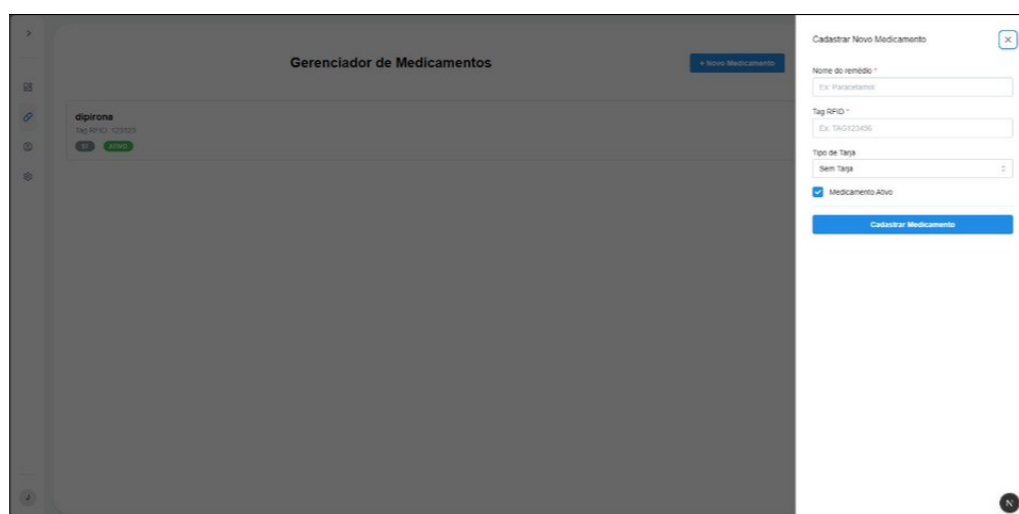


Figure 8. Medication management page.

Source: The authors.

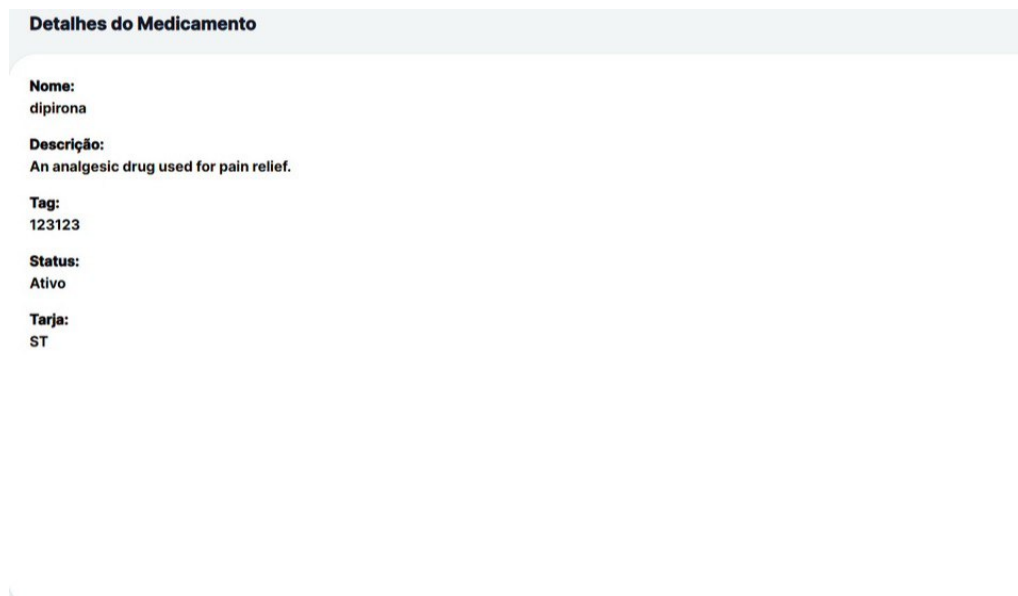


Figure 9. Medication details page.

Source: The authors.

The interface also allows users to view account details, enhancing control over which professionals are responsible for each movement, as shown in Figure 10. All data are securely managed and stored on the *backend*, while the *frontend*, built with *Next.js*, ensures dynamic updates of records and real-time display of movements and *statuses*.

A key highlight in the *hardware* results is the performance of the handcrafted copper coil antenna, which was able to maintain optimal frequency and signal intensity for reading low-cost passive tags. The performance of this component is crucial, as it underpins efficient communication between the tag and the *RFID* reader, ensuring that passage information is consistently captured and transmitted to the *backend* for real-time registration.

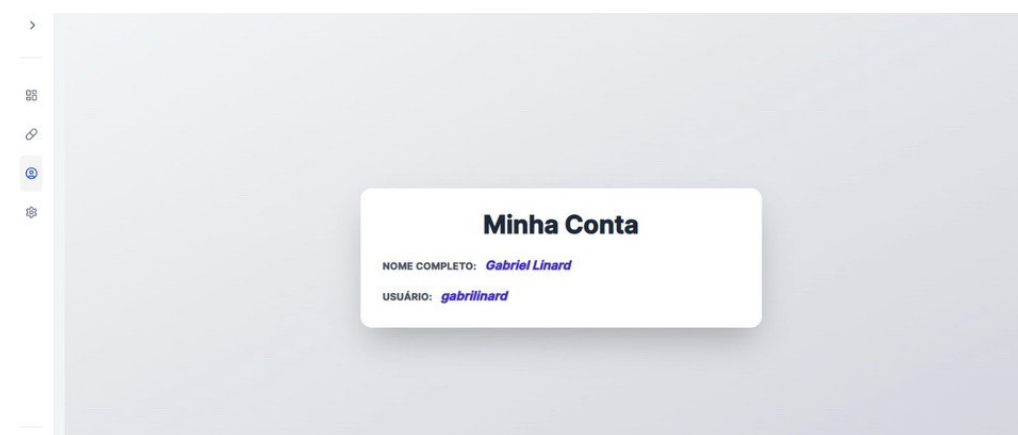


Figure 10. User account page.

Source: The authors.

As a practical outcome, the system enables efficient control over inventory, distribution traceability, and medication administration. This functionality reduces the occurrence of errors such as incorrect drug exchanges and the use of expired batches, strengthening information security and mitigating hospital losses, as highlighted by [Costa, Oliveira, & Santos, 2022] and [Ferreira, Lima, & Barbosa, 2023].

6. Discussion

6.1. Impact on Hospital Medication Management

The application of Radio-Frequency Identification (*RFID*) technologies in hospital environments has proven to be a promising solution for improving control, tracking, and security in medication management. Based on the simulations conducted in a virtual environment—using the *frontend* developed in *Next.js* with *React* and *TypeScript*, and the *backend* in *Django*—it was possible to verify that the integration between system components effectively met the requirements of traceability and monitoring. Among the main benefits observed, the system’s ability to reduce failures related to medication identification and administration stands out, since it ensures the association between the medication, the patient, and the storage location. This automation can mitigate common dispensing errors. Furthermore, the asynchronous communication model between the *hardware* device (*Arduino*) and the *backend*, through *HTTP* requests, demonstrated stable and reliable performance, with a low incidence of failures during data transmission.

6.2. Scalability and Replicability

The modular architectural model implemented favors the scalability of the solution. The clear separation between layers (*hardware*, *backend*, and *frontend*) allows the system to be adapted to different hospital units with minimal code adjustments. This structure facilitates project replication in other clinical and hospital contexts without the need for deep restructuring. The use of well-established and widely supported technologies—such as *Django* on the *backend* and *Next.js* with *React* on the *frontend*—provides ease of maintenance and ensures the potential for future expansions, such as integration with legacy hospital systems. Conducting tests in a simulated environment and continuously correcting errors identified during development allowed the anticipation of operational issues, validation of request integrity, and proper rendering of information on user interfaces, thereby enhancing the robustness of the solution.

6.3. Technological and Healthcare Contributions

This study provides a significant contribution to both the literature and the practice of hospital management by demonstrating that the combination of low-cost *hardware* (*Arduino* + *RFID* module), a robust *backend* (*Django* with *PostgreSQL*), and a modern *frontend* (*Next.js* with *React*) can offer an efficient, secure, and easily deployable solution for hospital medication control. The development process, focused on continuous testing and iterative corrections, ensured that the application met the essential requirements of security, traceability, and data integrity for clinical environments.

An important technological advancement lies in demonstrating the effectiveness of the handcrafted antenna for system miniaturization. Through experiments conducted

on a prototype simulating a hospital environment, it was shown that the necessary reading performance for passive tags can be achieved even in a reduced-size format. This dimensional scalability is crucial for the acceptance and deployment of *RFID* technology in hospital settings, where physical space is limited—thus supporting the feasibility of low-cost, large-scale tracking systems.

This technological integration contributes to the modernization of hospital supply-tracking processes and paves the way for future improvements, such as the inclusion of analytical *dashboards*, the generation of managerial reports, and the implementation of alert mechanisms for medication expiration or minimum stock control. Finally, the results reinforce the importance of customized and flexible solutions designed to meet the specific needs of each hospital environment, while respecting the infrastructural, connectivity, and regulatory particularities inherent to the healthcare sector.

7. Conclusion

The present study successfully achieved its objective by demonstrating the feasibility and effectiveness of developing an *RFID*-based medication traceability system, named “Passou Aqui,” for hospital logistics applications. The integration of low-cost *hardware*—composed of an *Arduino* microcontroller and an *RFID* MFRC522 module—with a robust *backend* built in *Django* and a dynamic *frontend* developed in *Next.js* proved to be an efficient and scalable architecture. The successful results obtained in simulated environments confirmed that the proposed solution is capable of providing real-time tracking, significantly reducing losses of high-value medical supplies, and optimizing inventory management—critical elements for the financial and operational sustainability of healthcare institutions.

One of the main contributions of the system lies in strengthening patient safety, as complete traceability and access control through authentication tokens and user profiles mitigate the risk of medication errors, such as incorrect drug administration or the use of expired batches. Such rigorous control is fundamental for compliance with national regulatory requirements, particularly those established by the **ANVISA (National Health Surveillance Agency)**, which demand greater transparency and oversight across the entire pharmaceutical supply chain. The detailed documentation of each stage in the medication’s movement provides an auditable record, essential for inspection processes and for the continuous improvement of internal protocols.

From a technological standpoint, this research validated the use of accessible *hardware* components whose performance results were enhanced by the development of a handcrafted antenna. As discussed, the miniaturization and efficiency of this antenna in maintaining the optimal reading frequency for passive tags demonstrate that *RFID* can be implemented at scale while respecting the spatial constraints typical of hospital environments. This advancement supports the replicability of the system in various healthcare contexts, positioning the solution as a viable and economically attractive alternative in the market.

Information security and data privacy constitute a central pillar of this solution. The entire *backend* was designed to manage and store movement data securely, employing encryption for data in transit and at rest. The implemented security architecture adheres

to the principles of the **General Data Protection Law (LGPD)**, particularly regarding the protection of personal data from logged-in operators and the assurance of integrity in both clinical and logistical records. Granular access control and detailed user activity logs ensure individual accountability for all operations, strengthening protection against fraud and data misuse.

For future work, it is suggested to expand the system by implementing predictive alert modules that employ artificial intelligence to identify anomalous patterns of deviation or excessive medication consumption. Additionally, direct integration with Electronic Health Record (EHR) systems and field testing in real hospital environments will be crucial to validate the system's operational scalability and usability on a large scale, consolidating "Passou Aqui" as a technological innovation serving healthcare and the efficient management of public and private resources.

8. Anexo

Github-Frontend. Github-backend. Github-Arduino.

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